AKTU • B.Tech CSE • EXAM PREPARATION

Machine Learning Important Questions

A beginner-first handbook: learn the concept from zero, understand the working, practice a small example, then write the exam answer.

9 focused topics

Approaches • Tom Mitchell • Issues • Linear & Logistic Regression • Data Science / AI / ML • EM • SVM • Concept Learning • Overfitting / Underfitting

How to Use This Handbook

THE LEARNING METHOD

1. What is it? → 2. Why do we need it? → 3. Core idea → 4. Working/steps → 5. Example → 6. Formula/diagram → 7. Exam-ready answer. This order is deliberately designed for a student learning ML for the first time.

Priority is given to the nine topics requested. Extra modern ML material is kept out unless it directly helps understand one of these topics.

Quick priority

Priority	Focus
Must prepare	Linear Regression, Logistic Regression, SVM, EM, Concept Learning, Tom Mitchell
Important	Approaches, Issues, AI/ML/Data Science, Overfitting/Underfitting
For numericals	Linear Regression, Logistic Regression probability, Find-S / Candidate-Elimination

UNIT I

1. Approaches to Machine Learning MUST PREPARE

WHAT IS THIS?

An ML approach tells us how a machine learns from information. The main distinction is the kind of feedback available: correct answers (supervised), no answers (unsupervised), a mixture (semi-supervised), or rewards/penalties (reinforcement).

Why do we need different approaches?

Real-world problems do not always provide the same kind of data. If we have labeled examples, supervised learning is natural. If labels are unavailable, unsupervised learning can search for structure. If labels are scarce, semi-supervised learning can use both labeled and unlabeled data. If an agent learns by interacting with an environment, reinforcement learning is appropriate.

1.1 Supervised Learning

The learner receives input–output pairs. It learns a mapping from input to a known target. Two major tasks are classification (categorical output) and regression (continuous numeric output).

Example: emails labeled spam/not-spam are used to learn a spam classifier.

1.2 Unsupervised Learning

The learner receives unlabeled data and tries to discover useful structure such as groups, associations, or lower-dimensional representations.

Example: grouping customers by purchasing behavior using clustering.

1.3 Semi-Supervised Learning

A small labeled set is combined with a large unlabeled set. This is useful when obtaining labels is expensive.

1.4 Reinforcement Learning

An agent interacts with an environment, chooses actions, and receives rewards or penalties. It learns a policy that aims to maximize cumulative reward.

Approach	Data / Feedback	Goal	Examples
Supervised	Labeled	Predict target	Linear Regression, Logistic Regression, SVM
Unsupervised	Unlabeled	Discover structure	K-Means, PCA
Semi-Supervised	Few labeled + many unlabeled	Use both sources	Self-training, label propagation
Reinforcement	Reward from interaction	Choose good actions	Q-Learning

REMEMBER

Labeled → Supervised | Unlabeled → Unsupervised | Mixed → Semi-Supervised | Reward → Reinforcement

EXAM-READY (5 MARKS)

Machine Learning approaches are classified by the feedback available during learning. Supervised learning uses labeled input-output pairs for classification or regression. Unsupervised learning uses unlabeled data to discover hidden patterns. Semi-supervised learning combines a small labeled set with a large unlabeled set. Reinforcement learning uses an agent-environment interaction in which actions receive rewards or penalties.

UNIT I

2. Tom Mitchell's Definition of Machine Learning MUST PREPARE

THE IDEA IN ONE LINE

A program learns when experience makes its measured performance on a task improve.

STANDARD DEFINITION

"A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E."

— Tom M. Mitchell, Machine Learning (1997)

Understand E, T and P

Symbol	Meaning	Question to ask
T — Task	What the program must do	What job is it performing?
P — Performance	How success is measured	How do we know it is doing well?
E — Experience	What it learns from	What data/interaction does it receive?

Worked example: spam classification

Element	Answer
T	Classify an email as spam or not-spam.
P	Classification accuracy (or another specified evaluation measure).
E	A collection of emails labeled spam/not-spam.

How to solve an E–T–P question

- Find the action/job → T.
- Find how success is judged → P.
- Find the data or past interaction used for learning → E.
- Check the key condition: performance measured by P improves with more E.

COMMON CONFUSION

E is not the result. E is the experience/data. T is not the algorithm. T is the task. P is not the training data. P is the performance measure.

EXAM-READY (5 MARKS)

According to Tom M. Mitchell, a program learns from experience E with respect to a task T and performance measure P when its performance at T, measured by P, improves with E. For spam classification, T is classifying emails as spam/not-spam, P is classification accuracy, and E is a labeled collection of past emails.

UNIT I

3. Issues / Challenges in Machine Learning IMPORTANT

WHAT IS THIS?

Even a good algorithm can perform badly if the data, model, evaluation process, or deployment environment is poor.

Core issues you should know

Issue	Why / Effect	Typical remedy
Insufficient data	Too little evidence → poor generalization	Collect more representative data; augmentation where appropriate
Noisy / poor-quality data	Wrong or unstable patterns	Cleaning, filtering, careful preprocessing
Missing values	Incomplete computations or bias	Suitable imputation or justified deletion
Irrelevant features	Adds noise and complexity	Feature selection / engineering
High dimensionality	Sparse data, overfitting, higher cost	Feature selection or dimensionality reduction
Overfitting	Very good training performance, poor unseen-data performance	Regularization, validation, simpler model
Underfitting	Model too simple; poor training and test performance	Better features or more suitable capacity
Class imbalance	Majority class dominates learning	Class weights / suitable resampling
Data leakage	Future/test information enters training → misleading performance	Strict separation and leakage-safe preprocessing
Computational cost	Large data/models need time and resources	Efficient algorithms, sampling, hardware
Interpretability	Complex models can be difficult to explain	Interpretable models / explanation methods
Changing data	Model can become outdated as distributions change	Monitoring and retraining

HOW TO WRITE THIS IN AN EXAM

Do not only list names. For 4–5 issues, use Problem → Why → Effect → Solution. This turns a list into an explanatory answer.

EXAM-READY (5–10 MARKS)

Major ML challenges include insufficient or poor-quality data, missing values, irrelevant features, high dimensionality, overfitting, underfitting, class imbalance, data leakage, computational cost, limited interpretability and changing data distributions. These issues can reduce generalization, reliability or deployment performance and are addressed through appropriate data preparation, model selection, validation, regularization, monitoring and retraining.

UNIT II

4. Linear Regression + Logistic Regression MUST PREPARE

Part A — Linear Regression

WHAT IS IT?

Linear Regression is a supervised regression algorithm used to predict a continuous numeric value. It assumes the target can be modeled as a linear combination of input features.

Simple intuition

Suppose x = hours studied and y = marks. Plot the points. Linear Regression finds a line that best represents the relationship, then uses the line to predict y for a new x.

CORE EQUATION

$\hat{y} = b_0 + b_1 x$

$\hat{y}$ = predicted value, b_0 = intercept, b_1 = slope, x = input feature.

Least Squares and error

A residual is the difference between an actual value and its prediction. Least Squares chooses the line that minimizes the squared residuals.

FORMULAS

$SSE = \Sigma(y_i - \hat{y}_i)^2$

$MSE = SSE / n$

$b_1 = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / \Sigma(x_i - \bar{x})^2$

$b_0 = \bar{y} - b_1\bar{x}$

Small numerical

WORKED EXAMPLE

Given x = [1,2,3,4,5], y = [2,4,5,4,5]. Then $\bar{x} = 3$ and $\bar{y} = 4$. $\Sigma(x_i - \bar{x})(y_i - \bar{y}) = 6$ and $\Sigma(x_i - \bar{x})^2 = 10$. Therefore $b_1 = 0.6$ and $b_0 = 2.2$. Regression line: $\hat{y} = 2.2 + 0.6x$. For x = 6, predicted y = 5.8.

Important assumptions

• Linearity: the mean relationship is adequately represented by a linear form.

• Independence: observations/errors are appropriately independent for the setting.

• Homoscedasticity: residual variance is approximately constant.

• Normality of residuals: mainly important for classical inference assumptions.

• Low multicollinearity: especially important in multiple regression.

Part B — Logistic Regression

WHAT IS IT?

Logistic Regression is a supervised classification algorithm, especially common for binary classification. It estimates a probability and then applies a decision rule to obtain a class.

Why not simply use Linear Regression?

A linear score can take any real value. A probability should lie between 0 and 1. Logistic Regression passes the linear score through the sigmoid function.

SIGMOID

$z = b_0 + b_1 x_1 + \ldots + b_n x_n$
$\sigma(z) = 1 / (1 + e^{-z})$
$P(y=1|x) = \sigma(z)$

Decision rule

With a 0.5 threshold: probability $\geq 0.5 \rightarrow$ class 1; probability $< 0.5 \rightarrow$ class 0. The threshold can be changed when the costs of different errors are different.

TINY NUMERICAL

If $z = 2$, then $\sigma(2) \approx 0.881$. So the estimated probability of class 1 is about 88.1%. With threshold 0.5, predict class 1.

LOSS FUNCTION

$J(\theta) = -(1/n)\Sigma[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)]$
This is binary cross-entropy / log loss.

Linear Regression	Logistic Regression
Predicts continuous numeric output	Predicts class probability / class
Straight-line output	Sigmoid-transformed output
Typical loss: squared error / MSE conventions	Typical loss: binary cross-entropy
Example: price / marks	Example: spam / not-spam

EXAM-READY

Linear: supervised regression using a linear equation and least squares to predict a continuous value. Logistic: supervised classification using a linear score passed through a sigmoid to estimate class probability, followed by a threshold-based decision. Know the equations, one numerical example, and the key difference between the two.

5. Data Science / AI / Machine Learning IMPORTANT

FIRST UNDERSTAND THE RELATIONSHIP

AI is the broad field concerned with intelligent behavior. ML is a major approach within AI in which systems learn patterns from data. Deep Learning is a subfield of ML based on multi-layer neural networks. Data Science is a broader data-centric discipline involving collection, cleaning, analysis, statistics, visualization and modeling.

Field	Main question / focus	Examples of work
Artificial Intelligence	How can a system perform an intelligent task?	Reasoning, planning, agents, perception
Machine Learning	How can a system learn patterns from data?	Classification, regression, prediction
Deep Learning	How can multi-layer neural networks learn representations?	Vision, speech, language
Data Science	How can we turn data into useful knowledge and decisions?	Cleaning, analysis, visualization, statistics, ML

COMMON MISTAKE

Do not write “Data Science is simply a subset of AI.” A safer statement is that Data Science and AI overlap, while ML is widely used in both predictive modeling and data-science workflows.

Easy mental model

Think of AI as the broad goal/field; ML as one important way to build systems that learn from data; DL as a neural-network-based family inside ML; and Data Science as the broader end-to-end practice of extracting useful information from data.

EXAM-READY (5 MARKS)

Artificial Intelligence is the broad field of intelligent systems. Machine Learning is a major approach within AI that learns patterns from data. Deep Learning is a subfield of ML using multi-layer neural networks. Data Science is an interdisciplinary, data-centric field involving data preparation, statistical analysis, visualization and predictive modeling. These areas overlap but are not identical.

6. Expectation-Maximization (EM) Algorithm MUST PREPARE

WHAT PROBLEM DOES EM SOLVE?

Sometimes data contains hidden (latent) variables. We cannot directly observe the hidden assignment, but we still want to estimate the model parameters. EM alternates between estimating the hidden part and improving the parameters.

Key vocabulary

Term	Meaning
Observed data	Values we can directly see.
Latent variable	Unknown/hidden quantity associated with the observed data.
Parameters	Values defining the probabilistic model.
Likelihood	How well parameter values explain the observed data.

The two steps

Step	What happens	Think of it as
E-step — Expectation	Using current parameters, estimate probabilities/expected values for hidden variables.	“Given the current model, what hidden assignments are likely?”
M-step — Maximization	Use those expected assignments to update parameters by maximizing the expected complete-data likelihood.	“Given those assignments, what model fits best?”

EM algorithm — step by step

- Initialize parameters.
- Perform E-step: estimate hidden-variable probabilities/expectations.
- Perform M-step: update parameters.
- Check convergence using likelihood change, parameter change, or a stopping rule.
- Repeat E-step and M-step until convergence or a maximum iteration limit.

INTUITIVE EXAMPLE

Imagine points generated by two overlapping groups, but the group label of each point is unknown. EM starts with rough group parameters. The E-step gives each point probabilities of belonging to the groups. The M-step recalculates the group parameters using those probabilities. Repeating the process refines the model.

IMPORTANT LIMITATION

EM can converge to a local optimum; therefore initialization can influence the final solution. Under standard EM assumptions, the likelihood is not decreased by an EM iteration.

REMEMBER

E → Estimate hidden variables → M → Maximize/update parameters → repeat

EXAM-READY (10 MARKS)

Expectation-Maximization is an iterative technique for parameter estimation when hidden variables or incomplete information are present. It starts with initial parameters, performs an E-step to estimate latent-variable expectations/probabilities, and performs an M-step to update parameters by maximizing the expected complete-data likelihood. These steps repeat until convergence. EM is useful in probabilistic mixture models and can depend on initialization.

7. Support Vector Machine (SVM) MUST PREPARE

WHAT IS IT?

SVM is a supervised learning algorithm used mainly for classification. It chooses a separating hyperplane that maximizes the margin between classes. The closest training points are the support vectors.

Start with the geometry

CORE EQUATIONS

$w \cdot x + b = 0 \rightarrow$ decision hyperplane
$w \cdot x + b = \pm 1 \rightarrow$ canonical margin boundaries
Margin width $= 2 / ||w||$

What do the terms mean?

Term	Simple meaning
Hyperplane	The decision boundary separating classes.
Margin	The safe gap between the decision boundary and the nearest points.
Support vectors	Training points closest to the boundary; they determine the margin/boundary.
C	Soft-margin penalty controlling the trade-off between margin size and violations.
Kernel	A function used to obtain nonlinear decision boundaries through an implicit feature-space mapping.

Hard margin vs soft margin

Hard Margin	Soft Margin
Assumes perfect separation.	Allows some margin violations / misclassification.
Sensitive to non-separable data and outliers.	More practical for noisy data.
Strict constraints.	Uses slack variables and penalty C.

Kernel trick

A kernel lets SVM work with nonlinear relationships by computing similarities corresponding to a higher-dimensional feature space without explicitly creating every transformed feature.

Kernel	Form / idea
Linear	$K(x,z) = x \cdot z$
Polynomial	$K(x,z) = (x \cdot z + c)^d$
RBF / Gaussian	$K(x,z) = \exp(-\gamma\|\|x-z\|\|^2)$

SOFT-MARGIN OBJECTIVE — CONCEPT

$$\text{minimize } \tfrac{1}{2}||w||^2 + C\Sigma\xi_i$$

The first term favors a large margin; the slack term penalizes violations.

EASY WAY TO REMEMBER

If many separating lines are possible, SVM prefers the one with the largest margin. The points closest to that line are the support vectors.

EXAM-READY (10 MARKS)

SVM is a supervised classification algorithm that finds a separating hyperplane with maximum margin. The closest training points are support vectors. Hard-margin SVM requires perfect separation, whereas soft-margin SVM allows violations using slack variables and C. For nonlinear data, kernel functions such as polynomial and RBF kernels can be used. SVM is especially useful in high-dimensional spaces but requires suitable parameter selection and feature scaling.

UNIT II

8. Concept Learning MUST PREPARE

WHAT IS CONCEPT LEARNING?

Concept Learning means learning a general rule from labeled examples. The learner searches a hypothesis space for rules that are consistent with the observed examples.

Basic terms

Term	Meaning
Instance	One training example described by attributes.
Concept	The target rule/classification to be learned.
Hypothesis	A candidate rule that predicts the target concept.
Hypothesis space H	The set of candidate hypotheses.
Version space	All hypotheses in H consistent with the training examples.

Find-S algorithm

Find-S begins with the most specific hypothesis and generalizes it only when a positive example requires a change. The basic Find-S procedure ignores negative examples.

• Start with the most specific hypothesis.

• Read each positive example.

• If a hypothesis attribute conflicts with the positive example, generalize that attribute.

• Ignore negative examples in basic Find-S.

• Return the resulting hypothesis.

SMALL FIND-S EXAMPLE

Attributes = <Sky, AirTemp, Humidity, Wind>.
Positive 1 = <Sunny, Warm, Normal, Strong> → h becomes <Sunny, Warm, Normal, Strong>.
Positive 2 = <Sunny, Warm, High, Strong> → Humidity differs, so generalize it: <Sunny, Warm, ?, Strong>.

Candidate-Elimination

Candidate-Elimination keeps the version space rather than only one hypothesis. It represents that space using two boundaries:

• S-boundary: maximally specific hypotheses consistent with the examples.

• G-boundary: maximally general hypotheses consistent with the examples.

Find-S	Candidate-Elimination
Returns one maximally specific consistent hypothesis.	Maintains the version space through S and G boundaries.
Basic procedure ignores negative examples.	Uses positive and negative examples to eliminate inconsistent hypotheses.
Simpler.	More complete but more computationally involved.

COMMON EXAM MISTAKE

Do not jump from the first training example directly to the final hypothesis. Show each example and the hypothesis after that example. This is especially important in Candidate-Elimination problems.

EXAM-READY (10 MARKS)

Concept Learning is the task of learning a target concept from labeled examples by searching a hypothesis space. Find-S starts with the most specific hypothesis and generalizes it using positive examples. Candidate-Elimination maintains the version space using the maximally specific S-boundary and maximally general G-boundary, updating these boundaries using positive and negative examples.

UNIT CROSS-CUTTING

9. Overfitting / Underfitting MUST PREPARE

THE BASIC PROBLEM

A model should learn the useful pattern, not merely memorize the training data. Underfitting means the model is too simple. Overfitting means the model is too closely fitted to training data, including noise.

Understand with training vs test performance

Situation	Training performance	Test / unseen performance	Likely diagnosis
Too simple	Poor	Poor	Underfitting
Appropriate complexity	Good	Good	Good generalization
Too complex	Very good	Much worse	Overfitting

Bias and variance

Bias is error associated with overly restrictive/simplified assumptions. Variance is sensitivity to the particular training sample. Underfitting is commonly associated with high bias; overfitting is commonly associated with high variance.

CONCEPTUAL DECOMPOSITION

Expected test error ≈ Bias^2 + Variance + Irreducible Noise

How to reduce overfitting

• Use more representative training data where possible.

• Reduce unnecessary model complexity.

• Use regularization such as L1/L2 when appropriate.

• Use validation/cross-validation for model selection and tuning.

• Use feature selection or dimensionality reduction when appropriate.

• Use suitable early stopping/dropout for neural-network settings.

How to reduce underfitting

• Increase model capacity when justified.

• Add informative features or improve representation.

• Reduce excessive regularization.

• Tune training and model settings appropriately.

QUICK DIAGNOSIS

Training accuracy 60%, test accuracy 58% → likely underfitting.
Training accuracy 99%, test accuracy 72% → classic warning sign of overfitting.
Exact diagnosis depends on the baseline and problem, but a large train–test gap is an important overfitting signal.

EXAM-READY (10 MARKS)

Underfitting occurs when a model is too simple to capture the underlying pattern, producing high training and test error. Overfitting occurs when a model fits training data excessively and performs poorly on unseen data. Underfitting is commonly associated with high bias and overfitting with high variance. The bias–variance trade-off seeks a model with good generalization. Regularization, validation, appropriate complexity and better data can help control these problems.

Final Revision: One-Page Recall

Topic	What you must be able to say/write
Approaches	Supervised, unsupervised, semi-supervised, reinforcement + comparison
Tom Mitchell	Definition + identify E, T, P from a scenario
Issues	At least 5 issues using Problem → Effect → Solution
Linear Regression	$\hat{y}=b_0+b_1x$ + least-squares formulas + numerical
Logistic Regression	Sigmoid + probability + threshold + cross-entropy
AI / ML / DS	Relationship and clear distinctions
EM	Latent variables + E-step + M-step + repeat
SVM	Hyperplane + margin + support vectors + soft margin + kernels
Concept Learning	Hypothesis space + Find-S + S/G boundaries
Over/Underfitting	Train/test behavior + bias/variance + remedies

High-yield questions

1. Explain the four approaches to Machine Learning with examples.
2. State Tom Mitchell's definition and identify E, T and P in a new problem.
3. Discuss important issues/challenges in Machine Learning.
4. Explain Linear Regression and solve a least-squares numerical.
5. Explain Logistic Regression with sigmoid, probability and threshold.
6. Differentiate Linear Regression and Logistic Regression.
7. Differentiate AI, ML and Data Science.
8. Explain EM with E-step and M-step.
9. Explain SVM with hyperplane, margin and support vectors.
10. Explain hard-margin and soft-margin SVM and the kernel trick.
11. Explain Concept Learning and Find-S.
12. Explain Candidate-Elimination with S and G boundaries.
13. Differentiate overfitting and underfitting and explain bias–variance trade-off.

Last 30-minute checklist

MEMORIZE

E–T–P • four learning approaches • Linear Regression equations • Logistic sigmoid • SVM margin/support vectors • E-step/M-step • Find-S • S/G boundaries • overfitting vs underfitting • bias–variance.

THE BEST WAY TO STUDY THIS HANDBOOK

Read the “What is this?” box → understand the example → close the page and explain it aloud → write the formula/steps from memory → attempt the exam-ready answer.